MIL–P–46678B (MU)
31 August, 1966
SUPERSEDING
MIL–P–46678A (MU)
12 April 1963

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, GAS, M426 ASSEMBLING, MARKING AND PACKING

1. SCOPE

1.1 This specification covers the assembly, marking and packing of one type of projectile designated as Projectile, 8 Inch, Gas, Persistent VX, M426, or designated as Projectile, 8 Inch, Gas Non-Persistent GB, M426.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 — Ammunition and Special Weapons; General Specification for.

MIL–I–45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes (ABC–STD–105).

MIL–STD–109 — Quality Assurance Terms and Definitions.

MIL–STD–1168 — Lot Numbering of Ammunition.

MIL–STD–1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

7548346 — Pallet for 8 Inch Projectile: Empty or Loaded.

8860620 — Projectile, 8 inch: Gas, M426.

8860621 — Projectile, 8 inch: Gas, Non-Persistent GB M426, Simulant Filled.

8860624 — Projectile, 8 inch: Gas, Persistent VX, M426 Simulant Filled.

PUBLICATIONS

ARMY

ET–8860620 — Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 **Assembly.** The assembly shall comply with all requirements specified on Drawing (dwg.) 8860620 and with all requirements specified in applicable specifications.

FSC 1320

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3.3 Proving ground.

3.3.1 *Premature burst.* There shall be no functioning of the projectile in the gun bore or in flight before impact, when tested as specified in 4.3.1.

3.3.2 *Functioning.* The projectile shall function with sufficient fragmentation to form a visible cloud, when tested as specified in 4.3.1.

3.4 Workmanship. The projectile shall be dry and free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define the terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall apply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all materials purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity

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shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot of projectiles shall contain:

(a) VX or GB loaded filling assemblies from one manufacturer in accordance with the same drawing, same drawing revision, same specification and specification revision of one loading lot interfix number.
(b) Projectile bursters from not more than one interfix lot from one manufacturer.
(c) Supplementary charges from not more than one interfix lot from one manufacturer.
(d) Burster projectile support assemblies from not more than one interfix lot from one manufacturer.

4.2.2 *Examination.* Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile burster, prior to assembly to projectile (see dwg. 8860620).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:			
1.	Cushioning pad missing	Visual	01001
2.	Closure missing	Visual	01002
Major: None defined			
Minor: None defined			

4.2.2.2 Loading assembly, prior to assembly of support (see dwg. 8860620).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Projectile burster inverted	Visual	02001
Major:	AQL 0.40 percent		
101.	Projectile burster missing	Visual	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	02003

4.2.2.3 Support, prior to assembly to projectile (see dwg. 8860630, a detail of dwg. 8860620).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Inside diameter	Gage	03001
202.	Total length, maximum	Gage	03002
203.	Outside diameter	Gage	03003

4.2.2.4 Fuze adapter, prior to assembly to projectile (see dwg. 8860620).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Coating missing from fuze adapter threads	Visual	04001
102.	O ring missing or damaged	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	04003

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4.2.2.5 Loading assembly, prior to assembly of fuze adapter (see dwg. 8860620).

Categories / Defects	Method of inspection	Code No.
Critical: None defined		
Major: AQL 0.40 percent		
101. Support missing	Visual	05001
102. Supplementary charge missing (when applicable at this stage of assembly)	Visual	05002
Minor: AQL 0.65 percent		
201. Evidence of poor workmanship (see 3.4)	Visual	05003

4.2.2.6 Loading assembly, prior to assembly of lifting plug (see dwg. 8860620).

Categories / Defects	Method of inspection	Code No.
Critical:		
1. Shell zone weight marking incorrect by two or more zones	Scale	06001
Major: AQL 1.0 percent		
101. Shell zone weight marking incorrect by one zone	Scale	06002
102. Length, max.	Gage	06003
103. Fuze adapter not seated and staked	Visual	06004
104. Spacer missing	Visual	06005
105. Supplementary charge missing (when applicable at this stage of assembly)	Visual	06006
Minor: AQL 0.65 percent		
201. Evidence of poor workmanship (see 3.4)	Visual	06007

4.2.2.7 Lifting plug, prior to assembly to projectile (see dwg. 8860620).

Categories / Defects	Method of inspection	Code No.
Critical: None defined		
Major: AQL 0.40 percent		
101. Corrosive preventive compound missing from threads	Visual	07001
102. Gasket missing or damaged	Visual	07002
Minor: AQL 0.65 percent		
201. Evidence of poor workmanship (see 3.4)	Visual	07003

4.2.2.8 Loading Assembly (see dwg. 8860620).

Categories / Defects	Method of inspection	Code No.
Critical:		
1. Marking misleading or unidentifiable including incorrect color code	Visual	08001
2. Projectile with incorrect type of gas (not as required for the particular lot being loaded)	Visual	08002
Major: AQL 0.65 percent		
101. Diameter of bourrelet (if shell is retouched)	Gage	08003
102. Torque test failure of lifting plug	Test	08004
103. Grommet missing	Visual	08005
104. Any weight zone punch mark missing or inadequate	Visual	08006
Minor: AQL 1.5 percent		
201. Total length	Gage	08007
202. Grommet obviously loose	Visual-	
203. Protective coating damaged (total bare area in excess of ¼ square inch	Manual	08008
204. Evidence of poor	Visual	08009
workmanship (see 3.4)	Visual	08010

4.2.2.9 Pallet, prior to strapping (see dwg. 7548346).

Categories / Defects	Method of inspection	Code No.
Critical: None defined		
Major: AQL 0.40 percent		
101. Cover assembly missing	Visual	09001
102. Unauthorized mixing of lots on same pallet	Visual	09002
Minor: AQL 1.0 percent		
201. Sleeve missing	Visual	09003
202. Incorrect number of projectile	Visual	09004

4.2.2.10 Pallet (see dwg. 7548346).

Categories / Defects	Method of inspection	Code No.
Critical: None defined		
Major: AQL 0.65 percent		
101. Board broken	Visual	10001
102. DOD symbol missing or incorrect	Visual	10002
103. Strapping missing or loose	Visual	10003
Minor: AQL 1.0 percent		
201. Marking misleading or unidentifiable	Visual	10004
202. Strapping improperly engaged	Visual	10005

4.2.3 Testing.

4.2.3.1 Premature burst (see 3.3.1) — Critical defect — Code No. 08011.

4.2.3.1.1 Beginning with the first lot produced and continuing until three consecutive

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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